

Physical AI SDK: 2026 Model Lifecycle & Delivery Strategy

What We're Delivering

Structured 4-Phase Model Lifecycle:



2026 Portfolio Targets:

- **20 production-ready models** across 6 categories
- **3 market segments:** Robotics, Healthcare, Industrial
- Unified SDK installer with integrated tooling

Model Categories:

Category	Models	Status
Vision/Detection	YoloV12, MobileSAM, CrossFormer	Active
VLMs	Qwen2.5-VL, Qwen3-VL, LLaMA3.2-Vision	Active
LLMs	DeepSeek-R1	In Progress
Speech	Whisper (ASR), XTTS (TTS)	Active
3D/Robotics	CenterPoint, OpenVLA	In Progress
Medical	MedSigLIP	In Progress

2026 Release Roadmap

R1: Foundation

- Lifecycle framework operational
- CI/CD automation & GitHub Pages/AMD docs
- Release logistics streamlined

R2: Scale

- **8 production-ready models** (P0/P1)
- Single SDK installer
- Quantization & Infra Beta (vLLM, ONNX-RT, Llama.cpp)
- Sample apps: ChatBot, Object Detection, GStreamer+ROCM

R3: Expand

- **8 additional models** (16 total)
- Enhanced quantization & profiling tools
- Robotics simulator integration
- Model deployment infrastructure

R4: Complete 1st cycle

- **20 total production models**
- Multi-segment deployment ready
- Model training infrastructure (Drafter)
- 2027 strategy planning

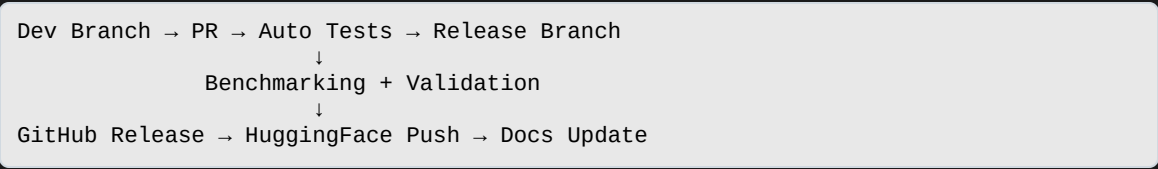
Open Source Distribution & Team Collaboration Model

GitHub + HuggingFace Release Strategy

What We Publish:

- Pre-optimized model weights (ONNX, TensorRT, ROCm)
- Quantized variants (INT8, FP16) for edge deployment
- Docker containers (GPU/NPU/Hybrid ready)
- Verified benchmark results & model cards

CI/CD Pipeline:



Infrastructure Tools:

- **vLLM** - LLM serving & inference
- **Llama.cpp** - Optimized CPU/GPU inference
- **ONNX Runtime** - Cross-platform deployment
- **Lemonade** - Performance profiling

Team Collaboration Model

PAVS (Internal) vs External Partners:

Stage	PAVS Team	External (MCW/AIG)
Enablement	Lead	Support
Profiling	Initial	Deep Analysis
Benchmarking	Setup	Automation
Fine-tuning	Goals+Lead	Execution

Handoff Criteria:

- Model runs on target hardware
- Profiling report complete
- Baseline benchmarks established
- Optimization opportunities explored

Key Benefits:

- Better ROI on external resources
- Clear deliverables with measurable outcomes
- Faster time to production
- PAVS Lead on goal-setting, partners focus on execution